Using deep learning to generate synthetic tax records

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What do we want?

Augmented tax unit microdata
for every year, including future years

What do we have?

- **One year of tax data, filers only**
  - About ten years out of date
  - Some values are blurred and some records are distorted
  - No non-filers, and none of the other variables a tax calculation needs: consumption, wealth, unrealized gains, estates
- **Surveys** that hold some of this information, but noisily and not at administrative quality
- **Aggregate targets**
  - Past years: IRS totals by AGI group, filing status and age range
  - Future years: CBO-style projections of 1040 items, or of total wages, business profits and similar totals

The task

Turn these scattered sources into

synthetic, realistic, augmented tax unit microdata

for every year

How has this been done before?

	Status	Weight	Wages	Interest	Capital gains
Filers (PUF)	Married	812	$142k	$310	$5.2k
	Single	1,404	$38k	$0	$0
	Married	655	$0	$4.9k	$61k
	Head of hh.	1,120	$51k	$12	$0
	Single	3	$2.3m	$88k	$3.9m

1. **Start with the IRS PUF**, whatever year is available.

Made-up records.

How has this been done before?

	Status	Weight	Wages	Interest	Capital gains
Billionaires	Married	1	$1.2m	$14m	$310m
Filers (PUF)	Married	812	$142k	$310	$5.2k
	Single	1,404	$38k	$0	$0
	Married	655	$0	$4.9k	$61k
	Head of hh.	1,120	$51k	$12	$0
	Single	3	$2.3m	$88k	$3.9m
Non-filers	Single	2,310	$0	$0	$0
	Single	1,870	$6.5k	$0	$0

2. **Add records**: non-filers, and the very rich the PUF leaves out.

Made-up records.

How has this been done before?

	Status	Weight	Wages	Interest	Capital gains	Net worth	Consump-tion	Tips
Billionaires	Married	1	$1.2m	$14m	$310m	$4.1b	$6m	$0
Filers (PUF)	Married	812	$142k	$310	$5.2k	$610k	$96k	$0
	Single	1,404	$38k	$0	$0	$9k	$34k	$6.1k
	Married	655	$0	$4.9k	$61k	$2.4m	$88k	$0
	Head of hh.	1,120	$51k	$12	$0	$31k	$47k	$0
	Single	3	$2.3m	$88k	$3.9m	$38m	$410k	$0
Non-filers	Single	2,310	$0	$0	$0	$2k	$19k	$0
	Single	1,870	$6.5k	$0	$0	$1k	$15k	$1.2k

3. Add columns: estimate Y given X in a survey, where X is also on the PUF; draw Y for each record.

Made-up records.

How has this been done before?

	Status	Weight	Wages	Interest	Capital gains	Net worth	Consump-tion	Tips
Billionaires	Married	1	$1.2m	$14m	$310m	$4.1b	$6m	$0
Filers (PUF)	Married	860 ↑	$142k	$310	$5.2k	$610k	$96k	$0
	Single	1,290 ↓	$38k	$0	$0	$9k	$34k	$6.1k
	Married	702 ↑	$0	$4.9k	$61k	$2.4m	$88k	$0
	Head of hh.	1,180 ↑	$51k	$12	$0	$31k	$47k	$0
	Single	2 ↓	$2.3m	$88k	$3.9m	$38m	$410k	$0
Non-filers	Single	2,450 ↑	$0	$0	$0	$2k	$19k	$0
	Single	1,760 ↓	$6.5k	$0	$0	$1k	$15k	$1.2k

4. **Reweight to the target year**: grow weights for population and dollars for prices, then move each weight up or down until the year's totals are hit.

Made-up records.

What were the problems with this approach?

- **It did not work well for past years.** The affordability project made this clear.
- **A separate model for each variable**, each estimating its conditional relationships in its own way.
- **Reweighting worked for only a few past years.** Tax law changed what the IRS targets measure, mainly TCJA cutting back itemized deductions.
- **Stretched further every year.** Plausible records for future years, but worse with each year away from 2015.

The ideal approach

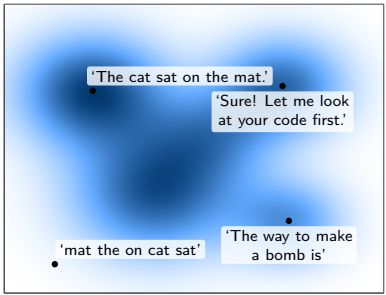
1. Take in survey data covering the joint relationships among every variable we might want to model.
2. Learn all of those relationships in **One Huge Joint Probability Distribution**.
3. For any target year, supply the targets that describe that year, and pull out of it the records that match them.

The result: a realistic synthetic microdata set for that year.

From a conversation with John and Harris.

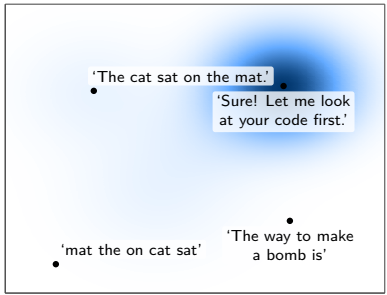
Language models: pretraining, then post-training

Pretraining



Read the whole internet and learn which sentences are likely: a prior over language

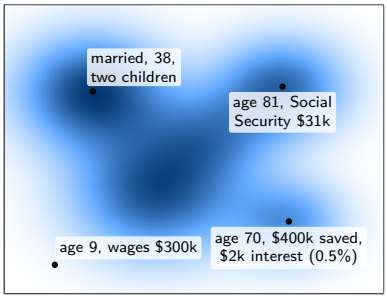
Post-training



Squeeze the mass toward a helpful, harmless assistant, and away from nonsense and harm

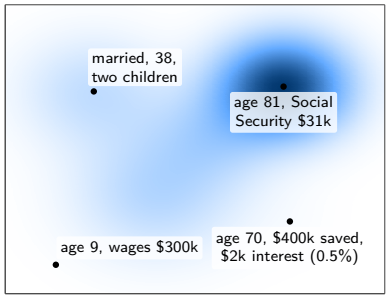
Our model: pretraining, then post-training

Pretraining



Learn which records are likely, from every source: a prior over records

Post-training for 2040



Squeeze the mass toward records that hit 2040's totals, each still coherent

Post-training finds the distribution closest to the prior that hits the year's totals.

The training data: about one million records from five sources

Source	Years	Records	What only it brings
PUF	2015	207,692	about 90 tax items
SCF	4 waves, 2013–2022	138,902	balance sheet (20 items), auto loans
CEX	6 years, 2015–2024	183,259	spending in 8 categories
CPS	5 years, 2015–2023	401,050	non-filers, children's ages, childcare
SIPP	5 years, 2015–2023	127,599	tips
All		1,058,502	

- Every source has married, dependents, the age range and wages.
- Every record carries its source and year as labels.
- Items a source never covers, like the balance sheet on the PUF, are left out rather than set to zero.

Surveys split into tax units by one set of CBO rules first.

Each dollar amount goes in as one of 84 ranges

- One range for exactly zero, 10 for losses, 73 for positive amounts.
- Cut at percentiles of the item's own weighted distribution, finer at the top.
- Wages on the 2015 PUF: every 2 points to p90, then every point to p99, every 0.1 to p99.9, every 0.01 to p99.99.

Wage range	Percentiles	PUF records
$0	(zero)	42,144
$1–1,320	0–2	2,772
$120,100–126,200	90–91	1,702
$4.73m–123.7m	99.99–100	3,511

source PUF	year 2015	married yes	dependents 2	age range 35–44	wages range 90–91	. . . about 90 items	home left out

Ranges and record counts: wages on the 2015 PUF. The row of boxes: an example PUF record as it goes in.

What the model learns, and what it returns

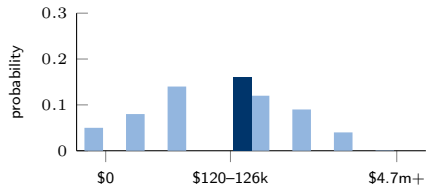
Training

1. Take a record. Hide a random share of its items, from one to all.
2. For each hidden item, the model gives a probability for each range.
3. Nudge its 1.4 million numbers to raise the probability of the true range.

80 passes over the records (2% held back to check): 74 minutes on one GPU. Two models, from different random starts.

What it returns

Asked "wages, given married, two dependents, aged 35–44, PUF, 2015, . . . ":



Draw a range: \$120,100–126,200. Pick a real 2015 PUF record in it, by weight: \$123,100. Grow with wages per return ($\times$1.117): **\$137,472** in 2019.

Bar heights made up; a few of the 84 ranges shown. The range and dollars: a real record of the 2019 file.

Generating the 2019 file

Which records

- 400,000 filers: the AGI group set first, the rest drawn given it.
- 54,379 non-filers, under the CPS label, in proportion to the CPS's expected non-filers.
- Half from each of the two models.

AGI group	Records	Returns each
below p50	101,838	720
p99–99.5	15,916	47
p99.99 and up	18,360	0.8

Each record, one item at a time

1. Tax items, in a random order.
2. Totals added up from their parts.
3. Exact ages, as the CPS would see the unit.
4. Balance sheet, as the SCF would see it, from one wave: 2019 (55%), 2022 (33%), 2016 (12%).
5. Spending, as the CEX would see it.
6. The CPS's person items; the SIPP's tips.

Each item: probabilities for its ranges, draw one, add it to what is known.

In 14% of filer records the drawn parts add up to another AGI group than the one set first. Reported, not forced.

One record from the 2019 file

Tax items (PUF label)

Joint return, dependents	yes, 2
Age range	35–44
Wages	$137,472
Taxable interest	$280
State refund	$1,076
Student loan interest	$246

People (CPS label)

Ages of the couple	41 and 49
Ages of the children	5 and 10
Wages of each	$59,128 and $78,344
Weeks worked	52 and 50

Balance sheet (SCF label, 2022 wave)

Home	$225,295
Mortgage	$139,061
Retirement accounts	$202,062
Deposits	$99,260
Net worth	$478,731

Spending (CEX label)

Total	$109,623
Food at home	$14,652

Weight (returns it stands for)

Drawn	565
After post-training	111

Nominal 2019 dollars. Selected items only: the record has others, such as hours worked, the other spending categories and auto loan interest.

Post-training: reweight to hit 753 targets

Targets for 2019	Number
SOI tables and business	577
CEX spending	104
SCF wealth, auto loans	52
W-2 tips	11
Non-filers	9

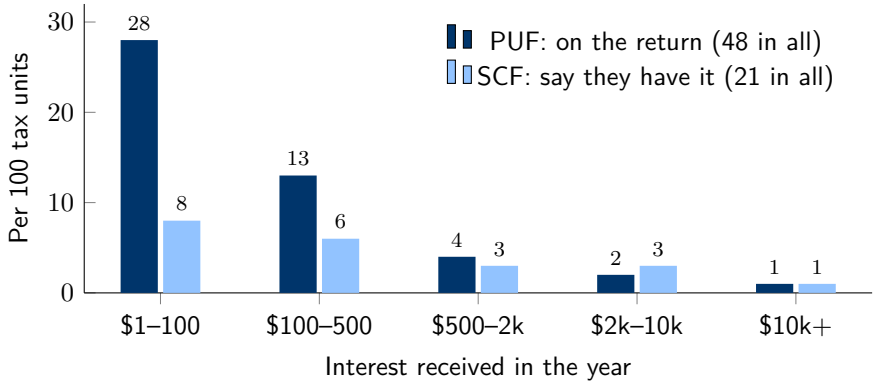
	Drawn	After	Target
Returns with no AGI (m)	3.25	2.16	2.13
Head of household returns (m)	23.86	21.59	21.66
AGI of the top 0.01% ($bn)	447	517	522
Joint returns with tips (m)	0.95	1.17	1.16
Units above SCF p99.9 net worth (%)	0.142	0.097	0.100
Vehicle spending per unit ($)	3,194	3,369	3,508

The solver changes only the weights, as little as it can, until every target is within one standard error of its value (half a standard error for the SCF's).

- All 753 targets met.
- Most weights end between 0.05 and 3.2 times where they started.
- Effective sample kept: 87%.

Drawn: scaled to SOI's returns by age and filing status. Most weights: 1st to 99th percentile. Effective sample: Kish's.

Challenge: surveys and returns measure differently



The gap is all in small amounts: they go unreported in the survey. Above $2,000 the two sources agree.

Married, wages $75–200k. 2015 PUF and 2016 SCF (tax year 2015), weighted.

The fix: two slots and a fill-in

SCF household	interest, as said	interest, as on a return
household 1	$250	$250
household 2	$0	?
household 3	$0	?
household 4	$1,800	$1,800

The answer given is kept. A second slot holds what the return would show, the only slot the PUF has.

Per 100 couples	PUF	SCF	Fill in
$1–100	28	8	20
$100–500	13	6	7
$500–2k	4	3	2
$2k and over	3	3	0
Any	48	21	28

The PUF sets how many are filled in, and at what sizes. The rest of the SCF record sets which households.

Married, wages $75–200k. 2015 PUF and 2016 SCF (tax year 2015), weighted. Rows may not add because of rounding.

Other challenges

- **Building tax units from surveys.** Surveys count households. Each one is split into tax units by the same rules for every survey, and then we decide who files.
- **The return itself changes.** TCJA cut back itemizing; new items, like the tips and overtime deductions, are on no past return.
- **A thin top.** Few very rich records in any source.
- **Future years.** No microdata, only a handful of projected totals.
- **Checking the result.** No source has every variable, so some relationships have nothing to check against.

Where we are

- A 2019 file: 400,000 filer records, plus non-filers
- Five sources: the 2015 PUF, the SCF, the CEX, the CPS and the SIPP, every survey split into tax units by the same rules
- Every PUF item, plus a balance sheet, consumption, tips and auto loan interest
- About 750 targets, all met within their standard errors: IRS tables by AGI, filing status and age, the top 1%, 0.1% and 0.01%, W-2 tips, SCF wealth

What is left

- Rules applied after drawing: credit eligibility, death probabilities, unrealized gains, the basis of capital gains
- The file Tax-Simulator reads, and a first run of Tax-Simulator on it
- A tax calculator inside post-training, for targets on computed tax
- Other years: 2015 to 2023, then the projection years
- The very top: the PUF's aggregate records and the Forbes 400

Questions